Retarded dispersion forces between molecules

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The frequency-dependent susceptibility of the molecules and of the electromagnetic field is used to calculate the force between two molecules at zero temperature. The field susceptibility at imaginary frequency, which is the Laplace transform of the retarded potentials, is found from the commutation relations, otherwise we do not need quantum field theory. A method of 'images' gives the force between a molecule and a conducting wall, and then we find the susceptibility of the field in the presence of the second molecule. This used to deduce the energy of the first one, and at long distances it turns out to be proportional to the mean square electric field produced by the second molecule. A reduced Hamiltonian involving only the electric and magnetic moments of the molecules gives the simplest proof and agrees with Casimir & Polder's theory (1948).

1. Introduction

The dispersion force between two molecules behaves quite differently according as their distance apart, R, is much less than or much greater than the wavelengths of the electronic absorption spectrum. At close distances the Coulomb interaction between the fluctuating electric dipole moments gives an energy proportional to $1/R^6$ (Eisenschitz & London 1930; London 1930), but when the distance is long the dipole moments fluctuate many times in the period a light signal takes to pass between the molecules, and retardation effects are important. Casimir & Polder (1948) showed that in this case the energy varies as $1/R^7$ and depends on the electric polarizabilities of the molecules. For example, for two like atoms of polarizability α the dispersion energy is

$$W = \frac{-23\hbar c}{4\pi}\frac{\alpha^2}{R^7}. \tag{1·1}$$

Their calculation was quite complicated, and has been followed by simpler ones. One approach uses a reduced Hamiltonian in which each atom is replaced by an electric dipole (Power & Zienau 1957; Aub, Power & Zienau 1957), and the most modern derivation uses the S-matrix of quantum electrodynamics (Dzyaloshinskii 1957). The interaction between individual molecules can also be derived as a special case of Lifschitz's theory of the attraction between macroscopic bodies (Lifschitz 1956; Landau & Lifschitz 1960), in which the force is deduced from the electromagnetic field fluctuations in the space between them (Dzyaloshinskii, Lifschitz & Pitaevskii 1961). In one sense the theory of molecular attractions is now complete, but there is still a need for an elementary discussion which gives better physical insight. The aim of this paper is to find the dispersion force between two molecules by a new method which uses elementary quantum mechanics and assumes almost no knowledge of quantum field theory.

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The proof depends on a new theorem which connects the interaction energy of two coupled quantum-mechanical systems with the response of each separate system to external perturbations (§ 3). The response depends in turn on the correlations between the fluctuations of certain physical quantities in the uncoupled systems (Kubo & Tomita 1954), and is described by a frequency-dependent susceptibility (§ 2). The quantized electromagnetic field may itself be regarded as having a certain susceptibility in response to particular kinds of perturbation (Callen & Welton 1951; Weber 1956), and the dispersion energy comes out as a function of the molecule and field susceptibilities at imaginary frequencies. To illustrate the basic ideas we calculate in § 4 the 'image force' between a molecule and a perfectly conducting metal wall, by considering the change in field susceptibility near the wall. The calculation becomes particularly simple if one uses a reduced Hamiltonian, derived in the appendix, and treats the electromagnetic field by the method of images. The force between two molecules is found in two stages. First, in § 5, we see how the field susceptibility is altered in the presence of the first molecule (A); then we find the change of energy when the second molecule (B) is coupled with the system (radiation and molecule A). The resulting expression (§ 6) is completely symmetrical between the two molecules, and brings out very clearly the physical significance of the long- and short-range energy dependences; in particular it shows why the force at long range depends on the static electric polarizability.

2. The generalized susceptibility

Suppose that a quantum-mechanical system is in its ground state. Then every physical quantity x characterizing the system will have a certain average value $\bar{x}$. (It is convenient to define x to be the difference between the quantity and its steady value, so that $\bar{x} = 0$.) When a small perturbation of the form

$$V = -\sum_i x_i f_i(t), \tag{2·1}$$

where x_i is the operator for the quantity x_i, and $f_i(t)$ is a given function of the time, acts on the system the mean value $\bar{x}_i(t)$ changes, and is a linear function of the values of the forces at previous times, of the form

$$\bar{x}_i(t) = \int_0^\infty K_{ik}(\tau) f_k(t-\tau)\, \mathrm{d}\tau. \tag{2·2}$$

(Here and in later equations a summation is made over repeated indices like k.) The 'response function' $K_{ik}(\tau)$ is characteristic of the system, and first-order perturbation theory (Kubo & Tomita 1954) leads to the important formula

$$K_{ik}(\tau) = (\mathrm{i}/\hbar)\langle[x_i(t+\tau), x_k(t)]\rangle. \tag{2·3}$$

The response function is thus the average value in the ground state of the commutator of $x_i(t+\tau)$ and $x_k(t)$, and is clearly independent of the time t. Here $x_i(t)$ is the Heisenberg operator of x_i in the unperturbed system.

When the forces are resolved into Fourier components $f_{k\omega}$ with time factors $\mathrm{e}^{-\mathrm{i}\omega t}$ the mean value of x_i has components

$$\bar{x}_{i\omega} = \alpha_{ik}(\omega) f_{k\omega}, \tag{2·4}$$

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where $\alpha_{ik}(\omega)$, the susceptibility, is the Fourier transform of $K_{ik}(\tau)$,

$$\alpha_{ik}(\omega) = \int_0^\infty K_{ik}(\tau)\, e^{i\omega\tau}\, d\tau. \tag{2·5}$$

Thus the susceptibility is connected through (2·3) with the correlations between the fluctuations of the quantities x_i and x_k at two different times. We refer the reader to Landau & Lifschitz's *Statistical physics* (1958) for a detailed discussion of the susceptibility, but it is useful to quote some results here.

When only one force acts, the real and imaginary parts of α are important quantities:

$$\alpha(\omega) = \alpha'(\omega) + i\alpha''(\omega). \tag{2·6}$$

For example, if f is an oscillating electric field and x is the dipole moment, α' gives the dispersion and α'' the rate of absorption of energy. In this case $K(\tau)$ is real and α' and α'' are respectively even and odd functions of the frequency. If several forces act it is useful to divide α_{ik} into 'real' and 'imaginary' parts with the properties

$$\alpha'_{ik}(\omega) = \alpha'_{ki}(-\omega), \quad \alpha''_{ik}(\omega) = -\alpha''_{ki}(-\omega). \tag{2·7}$$

Often the operators x_i are Hermitian, and then both the matrices α' and α'' are Hermitian. However, in the interaction of an atom with radiation it is useful to treat the current and charge densities, and the potentials, as four-vectors

$$J_\mu = (\mathbf{J}, i\rho), \quad A_\mu = (\mathbf{A}, i\phi) \tag{2·8}$$

with an imaginary time-component. In this case the matrices α' and α'' are still defined by (2·7) but are not Hermitian.

It is useful to have the susceptibilities of state n in terms of the matrix elements of x_i

$$\alpha'_{ik}(\omega) = \frac{P}{\hbar}\sum_m \left\{\frac{(x_i)_{nm}(x_k)_{mn}}{\omega_{mn}-\omega} + \frac{(x_k)_{nm}(x_i)_{mn}}{\omega_{mn}+\omega}\right\}, \tag{2·9}$$

$$\alpha''_{ik}(\omega) = (\pi/\hbar)\sum_m \{(x_i)_{nm}(x_k)_{mn}\,\delta(\omega+\omega_{nm}) - (x_k)_{nm}(x_i)_{mn}\,\delta(\omega+\omega_{mn})\}. \tag{2·10}$$

In (2·10) P stands for the principal value if $\omega = \pm\omega_{mn}$. We shall also need to know the energy shift

$$W = -\tfrac{1}{2}\alpha'_{ik}(0) f_{i0} f_{k0}, \tag{2·11}$$

caused by steady forces.

The well-known Kramers–Kronig relations and other important results are obtained by regarding the susceptibility $\alpha(\zeta)$ as an analytic function of the complex variable ζ, which may be defined by the time integrals

$$\alpha_{ik}(\zeta) = \int_0^{\pm\infty} K_{ik}(\tau)\, e^{i\zeta\tau}\, d\tau \tag{2·12}$$

above and below the real axis, respectively. The susceptibility has the symmetry property

$$\alpha_{ik}(\zeta) = \alpha_{ki}(-\zeta). \tag{2·13}$$

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There are no singularities of $\alpha_{ik}(\zeta)$ off the real axis, and it decreases like $1/\zeta$ when ζ is large. On the real axis there are poles at the transition frequencies $\pm\omega_{mn}$ and α tends to different limiting values as ζ tends to the real value ω from above or below.

$$\lim_{\epsilon\to 0}\alpha_{ik}(\omega\pm\mathrm{i}\epsilon)=\alpha'_{ik}(\omega)\pm\mathrm{i}\alpha''_{ik}(\omega) \tag{2·14}$$

and there is a discontinuity of $2\mathrm{i}\alpha''_{ik}(\omega)$ across the real axis. The ordinary susceptibility at real frequencies is now to be understood as the limit of $\alpha_{ik}(\omega+\mathrm{i}\epsilon)$ as one approaches the real axis from above. Equation (2·14) also leads to the important integral formula

$$\alpha_{ik}(\zeta)=\frac{1}{\pi}\int_{-\infty}^{+\infty}\frac{\alpha''_{ik}(\omega)}{\omega-\zeta}\,\mathrm{d}\omega. \tag{2·15}$$

At imaginary frequencies, $\omega=\mathrm{i}\xi$, the susceptibility is clearly the Laplace transform of the response function, according to (2·12), and has the following significance. Suppose the forces $f_k(t)$ increase exponentially from zero at $t=-\infty$. Then the mean value of x_i also increases exponentially and is given by

$$\bar{x}_i(t)=\alpha_{ik}(\mathrm{i}\xi)f_k\,\mathrm{e}^{\xi t};\quad f_k(t)=f_k\,\mathrm{e}^{\xi t}. \tag{2·16}$$

It is clear from (2·16) that $\alpha_{ik}(\mathrm{i}\xi)$ must be real if the operators x_i are Hermitian.

To calculate the long-range force between molecules one needs the susceptibility of both the molecules and the electromagnetic field (Weber 1956; Källén 1958) to certain forces. For the field take each variable x_i to be a component $A_\mu(\mathbf{r})$ of the vector potential at a certain point $\mathbf{r}$, and x_k to be a component $A_\nu(\mathbf{r}')$ at some other point $\mathbf{r}'$. We study the change in the average value $\bar{A}_\mu(\mathbf{r},t)$ caused by a perturbation

$$V=-\int A_\mu(\mathbf{r})j_\mu(\mathbf{r},t)\,\mathrm{d}\mathbf{r}, \tag{2·17}$$

in which $j_\mu(\mathbf{r},t)$ are given functions of time, corresponding to a certain distribution of external charges and currents, and $A_\mu(\mathbf{r})$ is the four-vector $(\mathbf{A},\mathrm{i}\phi)$. The response function is then

$$K_{\mu\nu}(\mathbf{r},\mathbf{r}';\tau)=(\mathrm{i}/\hbar)\langle[A_\mu(\mathbf{r},\tau),A_\nu(\mathbf{r}',0)]\rangle \tag{2·18}$$

and the well-known commutation relations of the potentials (Dirac 1958) show that

$$K_{\mu\nu}(\mathbf{r},\mathbf{r}';\tau)=\frac{c}{R}\{\delta(R-c\tau)-\delta(R+c\tau)\}\delta_{\mu\nu}, \tag{2·19}$$

(here $R=|\mathbf{r}-\mathbf{r}'|$) so that the change in the mean value of A is just the retarded potential produced by the external currents j. The corresponding susceptibility is written $A_{\mu\nu}(\mathbf{r},\mathbf{r}';\omega)$ and has the value

$$A_{\mu\nu}(\mathbf{r},\mathbf{r}';\omega)=\delta_{\mu\nu}\,\mathrm{e}^{\pm\mathrm{i}kR}/R \tag{2·20}$$

above and below the real axis, where $\omega=ck$. On the imaginary axis, where $\xi=ck$ the susceptibility decreases exponentially in both directions

$$A_{\mu\nu}(\mathbf{r},\mathbf{r}';\mathrm{i}\xi)=\delta_{\mu\nu}\,\mathrm{e}^{-kR}/R. \tag{2·21}$$

Another important perturbation is of the type

$$V=-\int\{\mathbf{E}(\mathbf{r}).\mathbf{p}(\mathbf{r},t)+\mathbf{H}(\mathbf{r}).\mathbf{m}(\mathbf{r},t)\}\,\mathrm{d}\mathbf{r}, \tag{2·22}$$

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in which the external force is caused by a given set of electric and magnetic dipoles in space. Then the response functions which govern the changes of $\bar{\mathbf{E}}$ and $\bar{\mathbf{H}}$ must be found from the commutation relations of these fields. For example the response function

$$E_{ik}(\mathbf{r},\mathbf{r}';t-t') = (\mathrm{i}/\hbar)\langle[E_i(\mathbf{r},t), E_k(\mathbf{r}',t')]\rangle \tag{2·23}$$

and the corresponding susceptibility $E_{ik}(\mathbf{r},\mathbf{r}';\omega)$ give the average electric field

$$\bar{E}_i(\mathbf{r})_\omega = E_{ik}(\mathbf{r},\mathbf{r}';\omega)\, p_k(\mathbf{r}')_\omega \tag{2·24}$$

produced at $\mathbf{r}$ by an oscillating electric dipole at $\mathbf{r}'$. There will also be susceptibilities which connect $\mathbf{E}$ with $\mathbf{H}$ and $\mathbf{H}$ with $\mathbf{H}$. The value of E_{ik} is easily found from (2·18) (Heitler 1954) if we remember that $\mathbf{E} = -\nabla\phi - (1/c)\dot{A}$:

$$E_{ik}(\mathbf{r},\mathbf{r}';t-t') = \left[-\nabla_i\nabla'_k + \frac{1}{c^2}\frac{\partial^2}{\partial t\,\partial t'}\delta_{ik}\right]\frac{c}{R}\{\delta(R-c\tau)-\delta(R+c\tau)\}, \tag{2·25}$$

and then the susceptibility comes out as

$$E_{ik}(\mathbf{r},\mathbf{r}';\omega) = \{\nabla_i\nabla_k - \delta_{ik}\nabla^2\}\mathrm{e}^{\pm\mathrm{i}kR}/R. \tag{2·26}$$

Finally, we study the currents produced in a molecule by a perturbation

$$V = -\int J_\mu(\mathbf{r})\, a_\mu(\mathbf{r},t)\,\mathrm{d}\mathbf{r} \tag{2·27}$$

which is the relativistic interaction energy of an electron with a certain external classical field. The changes of the current are described by the following susceptibility

$$\alpha_{\mu\nu}(\mathbf{r},\mathbf{r}';\zeta) = \frac{1}{\hbar}\sum_m\left\{\frac{J^{nm}_\mu(\mathbf{r})\,J^{mn}_\nu(\mathbf{r}')}{\omega_{mn}-\zeta} + \frac{J^{nm}_\nu(\mathbf{r}')\,J^{mn}_\mu(\mathbf{r})}{\omega_{mn}+\zeta}\right\}. \tag{2·28}$$

In the non-relativistic limit the operator V and the susceptibility are rather complicated expressions which we shall not take up here.

3. Mutual polarization energy of two systems

Suppose that two quantum-mechanical systems A and B in their ground states are weakly coupled together, with an interaction Hamiltonian

$$V = -\sum_i x_i y_i, \tag{3·1}$$

where x_i and y_i are physical quantities respectively belonging to A and B, and the average value of each quantity vanishes in the ground state of the uncoupled system. The energy shift W will be of the second order in V, and we now prove that it is related to the susceptibilities $\alpha_{ik}(\mathrm{i}\xi)$ and $\beta_{ik}(\mathrm{i}\xi)$ of the two systems at imaginary frequencies by the following equation

$$W = -\frac{\hbar}{2\pi}\int_0^\infty \alpha_{ik}(\mathrm{i}\xi)\,\beta_{ki}(\mathrm{i}\xi)\,\mathrm{d}\xi. \tag{3·2}$$

We start from the usual perturbation formula

$$W = -\sum_{a,b}{}' \frac{(x_i)^{0a}(y_i)^{0b}\,(x_k)^{a0}\,(y_k)^{b0}}{\hbar(\omega_{a0}+\omega_{b0})}, \tag{3·3}$$

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in which a and b denote the excited states of A and B, while the sum is over states in which both systems are excited at once. As the excitation frequencies of the ground state are all positive one can use (2·10) to write W in terms of the 'imaginary parts' of the susceptibilities

$$W = -\frac{\hbar}{\pi^2}\int_0^\infty \mathrm{d}\omega_1 \int_0^\infty \mathrm{d}\omega_2 \frac{\alpha''_{ik}(\omega_1)\,\beta''_{ik}(\omega_2)}{\omega_1+\omega_2}. \tag{3·4}$$

Now it is possible to treat the frequencies ω_1 and ω_2 as complex variables and the integrals in (3·4) as contour integrals. From (2·14) we obtain

$$W = -\frac{\hbar}{2\pi^2 \mathrm{i}}\int_0^\infty \mathrm{d}\omega_1 \int_0^\infty \mathrm{d}\omega_2 \frac{\alpha''_{ik}(\omega_1)\{\beta_{ik}(\omega_2+\mathrm{i}\epsilon)-\beta_{ik}(\omega_2-\mathrm{i}\epsilon)\}}{\omega_1+\omega_2}. \tag{3·5}$$

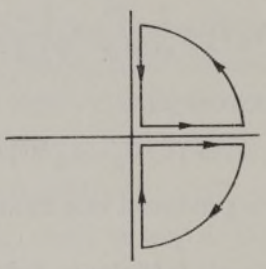


FIGURE 1

The contours $\zeta_2 = (\omega_2 \pm \mathrm{i}\epsilon)$ are completed round the upper and lower quadrants of figure 1, and give an integral along the whole imaginary axis

$$W = -\frac{\hbar}{2\pi^2}\int_0^\infty \mathrm{d}\omega_1 \int_{-\infty}^{+\infty} \mathrm{d}\xi \frac{\alpha''_{ik}(\omega_1)\,\beta_{ik}(\mathrm{i}\xi)}{\omega_1+\mathrm{i}\xi}. \tag{3·6}$$

Since $\beta_{ik}(-\mathrm{i}\xi) = \beta_{ki}(+\mathrm{i}\xi)$ the integral can be taken along the positive imaginary axis only

$$W = -\frac{\hbar}{2\pi^2}\int_0^\infty \mathrm{d}\omega_1 \int_0^\infty \mathrm{d}\xi \alpha''_{ik}(\omega_1)\left\{\frac{\beta_{ik}(\mathrm{i}\xi)}{\mathrm{i}\xi+\omega_1}-\frac{\beta_{ki}(\mathrm{i}\xi)}{\mathrm{i}\xi-\omega_1}\right\}. \tag{3·7}$$

Finally, one replaces $\alpha''_{ik}(\omega_1)$ in the second fraction by $-\alpha''_{ki}(-\omega_1)$ and combines the two fractions into a single ω_1 integral from $-\infty$ to $+\infty$, which (2·15) now shows is just $\pi\alpha_{ik}(-\mathrm{i}\xi)$ or $\pi\alpha_{ki}(+\mathrm{i}\xi)$. Thus we obtain (3·2).

The result is valid even when the operators x_i and y_i are not Hermitian. However, if they are both Hermitian and real the proof is very easy. We substitute (2·9), with ω replaced by $\mathrm{i}\xi$ into (3·2), which becomes

$$-\frac{\hbar}{2\pi}\sum'_{a,b}\int_0^\infty \mathrm{d}\xi \frac{2\omega_{a0}(x_i)^{0a}(x_k)^{a0}}{\hbar(\omega_{a0}^2+\xi^2)}\,\frac{2\omega_{b0}(y_k)^{0b}(y_i)^{b0}}{\hbar(\omega_{b0}^2+\xi^2)} \tag{3·8}$$

and use the definite integral

$$\int_0^\infty \mathrm{d}x \frac{ab}{(a^2+x^2)(b^2+x^2)} = \frac{\pi}{2(a+b)} \tag{3·9}$$

to arrive back at (3·3).

The susceptibilities $\alpha(\mathrm{i}\xi)$ and $\beta(\mathrm{i}\xi)$ both decrease to zero as ξ tends to infinity. It may, however, happen that $\alpha(\mathrm{i}\xi)$ decreases very rapidly with ξ while $\beta(\mathrm{i}\xi)$ remains

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sensibly constant. If β is constant it means that the response function of B is equivalent to a delta-function $\delta(\tau)$ as far as it affects A; that is, the system A changes so slowly that the response of B may be considered instantaneous. We shall call this situation the 'quasi-static case for B'. One can now replace $\beta(\mathrm{i}\xi)$ by its value at zero frequency, and get the energy shift

$$W = -\frac{\hbar}{2\pi}\beta_{ik}(0)\int_0^\infty \alpha_{ki}(i\xi)\,\mathrm{d}\xi. \tag{3·10}$$

The ground state of B is not degenerate, so that $\beta_{ik}(0) = \beta'_{ik}(0)$ and $\beta'_{ik}(0) = \beta'_{ki}(0)$. We may therefore use (2·13) to replace $\alpha_{ki}(\mathrm{i}\xi)$ in (3·10) by $\frac{1}{2}[\alpha_{ik}(\mathrm{i}\xi)+\alpha_{ik}(-\mathrm{i}\xi)]$. Integration round the contour of figure 1 leads back to

$$\int_0^\infty \alpha''_{ik}(\omega)\,\mathrm{d}\omega = \frac{\pi}{\hbar}\sum_a{}'(x_i)^{0a}(x_k)^{a0} = \frac{\pi}{\hbar}\langle x_i x_k\rangle, \tag{3·11}$$

which is just the mean value of $x_i x_k$, as follows from (2·10). The energy shift

$$W = -\tfrac{1}{2}\beta'_{ik}(0)\langle x_k x_i\rangle, \tag{3·12}$$

is thus connected with the time average of the fluctuations of system A, and behaves as if A were the source of slowly varying random forces proportional to the quantities x_i (compare (3·12) with (2·11)).

4. The force between a molecule and a conducting wall

When a molecule in free space interacts with the zero-point vibrations of the radiation field there is a certain energy change, the self-energy of the molecule. Near a perfectly conducting metal wall these fluctuations of the field are altered. The self-energy changes and causes a force between the molecule and the wall. Casimir & Polder (1948) were the first to calculate this force. They treated the radiation as a set of standing waves in a large metal box, and calculated the second-order energy shift in the presence of an atom. They found that the force behaved like the force between an oscillating dipole and its electrical image in the wall, and at long distances it was proportional to the static polarizability of the atom.

We shall calculate the force directly from the frequency-dependent susceptibilities $\alpha_{\mu\nu}(\mathbf{r},\mathbf{r}';\mathrm{i}\xi)$ and $A_{\nu\mu}(\mathbf{r}',\mathbf{r};\mathrm{i}\xi)$ of the molecule and the radiation in the presence of the wall. It is evident that the mutual polarization energy (3·2) due to the wall depends only on the difference $\tilde{A}$ between the susceptibility of the field near the wall and in free space

$$W = -\frac{\hbar}{2\pi}\int_0^\infty \mathrm{d}\xi \iint \alpha_{\mu\nu}(\mathbf{r},\mathbf{r}';\mathrm{i}\xi)\,\tilde{A}_{\nu\mu}(\mathbf{r}',\mathbf{r};\mathrm{i}\xi)\,\mathrm{d}\mathbf{r}\,\mathrm{d}\mathbf{r}'. \tag{4·1}$$

The most elegant way to find $\tilde{A}$ is to remember that the wave arriving at $\mathbf{r}'$ is the sum of a direct wave from $\mathbf{r}$ and a reflected wave which emanates from $\tilde{\mathbf{r}}$, the image of $\mathbf{r}$ in the wall. The quantized field operators $A_\mu(\mathbf{r},\tau)$ in equation (2·18) obey Maxwell's equations and satisfy the usual boundary conditions at the wall (Schweber 1961). One is thus able to find a solution by the method of images. To be definite, let the wall be the plane $x = 0$, and the centre of the molecule be fixed at $(\mathrm{X}, 0, 0)$. Then each point $\mathbf{r} = (x, y, z)$ has an image $\tilde{\mathbf{r}} = (-x, y, z)$, and one finds that

$$\tilde{A}_{\mu\nu}(\mathbf{r}',\mathbf{r};\mathrm{i}\xi) = \pm A_{\mu\nu}(\mathbf{r}',\tilde{\mathbf{r}};\mathrm{i}\xi) = \pm\delta_{\mu\nu}\mathrm{e}^{-kR}/R, \tag{4·2}$$

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where R is the distance from $\mathbf{r}'$ to $\tilde{\mathbf{r}}$ and the sign is $+$ for $\mu = 1$, $-$ for $\mu = 2, 3, 4$. Apart from the sign $\tilde{A}$ is the susceptibility of free space between $\mathbf{r}'$ and the image of $\mathbf{r}$. The corresponding energy shift is

$$W = -\frac{\hbar}{2\pi}\sum_{\mu} \pm \int_0^\infty \mathrm{d}\xi \iint \alpha_{\mu\mu}(\mathbf{r}, \tilde{\mathbf{r}}'; \mathrm{i}\xi)\frac{\mathrm{e}^{-kR}}{R}\,\mathrm{d}\mathbf{r}\,\mathrm{d}\mathbf{r}', \tag{4·3}$$

where $\xi = ck$, and we could now proceed by reducing α in (2·28) to its non-relativistic form. This is rather complicated and another method is far simpler.

In the appendix we show that the usual Hamiltonian $V = -\int J_\mu(\mathbf{r})\, A_\mu(\mathbf{r})\,\mathrm{d}\mathbf{r}$ for the interaction between the radiation field and the electrons in a neutral molecule is equivalent to the reduced Hamiltonian (Power & Zienau 1957)

$$V = -\int \{\mathbf{P}(\mathbf{r}).\mathbf{E}(\mathbf{r}) + \mathbf{M}(\mathbf{r}).\mathbf{H}(\mathbf{r})\}\,\mathrm{d}\mathbf{r}. \tag{4·4}$$

Here $\mathbf{P}(\mathbf{r})$ is the electric polarization at $\mathbf{r}$ and $\mathbf{M}(\mathbf{r})$ is the magnetic moment of the orbital and spin currents. We shall neglect the magnetic moments, though they can be included (Mavroyannis & Stephen 1962), and replace the polarization $\mathbf{P}(\mathbf{r})$ by a point dipole at $\mathbf{x}$, the centre of the molecule. This step is only justified if the size of the molecule is small compared with the important photon wavelengths. The effective interaction reduces to

$$V = -\mathbf{P}.\mathbf{E}(\mathbf{x}) \tag{4·5}$$

which is just the type of Hamiltonian for which the susceptibility of the electric field is useful. The mutual polarization energy due to the wall is clearly

$$W = -\frac{\hbar}{2\pi}\int_0^\infty \alpha_{ik}(\mathrm{i}\xi)\,\tilde{E}_{ki}(\mathbf{x}, \mathbf{x}; \mathrm{i}\xi)\,\mathrm{d}\xi, \tag{4·6}$$

where $\alpha_{ik}(\mathrm{i}\xi)$ is the ordinary electric polarizability tensor of the molecule, and $\tilde{E}_{ki}$ is the change of electric field susceptibility produced by the wall. It is easy to derive $\tilde{E}_{ki}$ from $\tilde{A}_{\mu\nu}$ which we have already found by the method of images

$$\tilde{E}_{ki}(\mathbf{x}, \mathbf{x}; \mathrm{i}\xi) = \pm E_{ki}(\tilde{\mathbf{x}}, \mathbf{x}; \mathrm{i}\xi). \tag{4·7}$$

Here $E_{ki}(\tilde{\mathbf{x}}, \mathbf{x}; \mathrm{i}\xi)$ is the susceptibility of free space between x and its image. The sign is $+$ for $k = x$, $-$ for y, z, and independent of i. To interpret (4·7) it is useful to imagine that the real molecule is accompanied by an electrical image molecule, centred at $(-X, 0, 0)$ whose charges are reversed in sign (figure 2). In the image molecule the dipole moment $\tilde{\mathbf{p}}$ at $\tilde{\mathbf{r}}$ bears the relation

$$(\tilde{p}_x, \tilde{p}_y, \tilde{p}_z) = (+p_x, -p_y, -p_z) \tag{4·8}$$

to the moment of the real molecule. Then if we extend the definition of the electric polarizability so that $\tilde{\alpha}_{ik}(\mathrm{i}\xi)$ gives the change in the image dipole $\tilde{p}_i$ produced by a field E_k, equation (4·6) becomes

$$W = -\frac{\hbar}{2\pi}\int_0^\infty \tilde{\alpha}_{ik}(\mathrm{i}\xi)\,E_{ki}(\tilde{\mathbf{x}}, \mathbf{x}; \mathrm{i}\xi)\,\mathrm{d}\xi. \tag{4·9}$$

The energy can therefore be written to look like the mutual interaction of the fluctuating molecular dipole and its image via the zero-point vibrations of the electric field.

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E_{ki} decreases exponentially with ξ according to (2·26)

$$E_{ki}(\tilde{\mathbf{x}}, \mathbf{x}; \mathrm{i}\xi) = [\nabla_k \nabla_i - \delta_{ki} \nabla^2] \frac{\mathrm{e}^{-kR}}{R} \tag{4·10}$$

and limiting values of W at long and short distances follow at once as examples of the 'quasi-static' case (3·10). At close distances $\tilde{\alpha}_{ik}(\mathrm{i}\xi)$ decreases much faster than E_{ki}, because a light wave can travel from $\mathbf{x}$ to $\tilde{\mathbf{x}}$ many times in the period of one electronic fluctuation. Thus

$$\begin{aligned} W &= -\frac{\hbar}{2\pi} E_{ki}(\tilde{\mathbf{x}}, \mathbf{x}; 0) \int_0^\infty \tilde{\alpha}_{ik}(\mathrm{i}\xi)\, \mathrm{d}\xi \\ &= -\tfrac{1}{2} \langle p_i \tilde{p}_k \rangle E_{ki}(\tilde{\mathbf{x}}, \mathbf{x}; 0) \\ &= \tfrac{1}{2} \langle (\mathbf{p}.\nabla)\, (\mathbf{p}.\nabla')\, 1/R \rangle. \end{aligned} \tag{4·11}$$

This is just half the mean Coulomb energy of the fluctuating dipole moment of the ground state in the presence of its image, and has the value

$$W = -\tfrac{1}{2} \langle 2p_x^2 + p_y^2 + p_z^2 \rangle / R^3. \tag{4·12}$$

To obtain (4·11) one puts $k = 0$ in (4·10) and notices that $\nabla^2(1/R) = 0$.

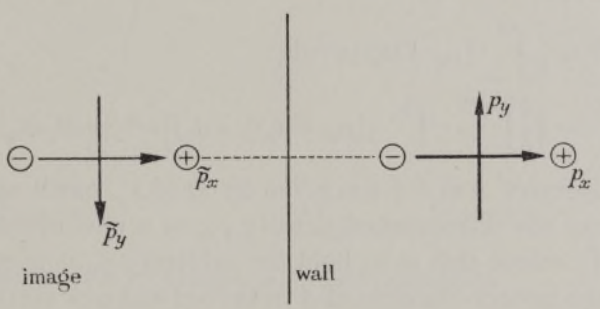


FIGURE 2

In the long-distance limit the exponential decay of E_{ik} is important. The energy

$$\begin{aligned} W &= -\frac{\hbar}{2\pi} \tilde{\alpha}_{ik}(0) \int_0^\infty E_{ki}(\tilde{\mathbf{x}}, \mathbf{x}; \mathrm{i}\xi)\, \mathrm{d}\xi \\ &= -\tfrac{1}{2} \tilde{\alpha}'_{ik}(0)\, \langle E_k(\tilde{\mathbf{x}})\, E_i(\mathbf{x}) \rangle \end{aligned} \tag{4·13}$$

behaves as if the dipole and its image were subject to a low-frequency random field, and depends on the free-space fields at $\mathbf{x}$ and $\tilde{\mathbf{x}}$. The integral in (4·13) is found by substituting c/R for e^{-kR} in (4·10), and gives Casimir & Polder's result

$$W = -(2\hbar c/\pi R^4) \{\alpha_{xx} + \alpha_{yy} + \alpha_{zz}\}. \tag{4·14}$$

The components of $E_{ik}(\tilde{\mathbf{x}}, \mathbf{x}; \mathrm{i}\xi)$ will be needed later. Those that do not vanish are E_{xx}, and $E_{yy} = E_{zz}$;

$$E_{xx} = (2/R^3)\, (1 + kR)\, \mathrm{e}^{-kR}; \quad E_{yy} = -(1/R^3)\, (1 + kR + k^2R^2)\, \mathrm{e}^{-kR}. \tag{4·15}$$

From them one may calculate the force at intermediate distances.

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5. Field susceptibility changed by a molecule

The interaction between a molecule and the radiation field alters the susceptibility of the field at other parts of space, and thereby produces a force between the molecule and other bodies. We therefore consider the response function $\hat{K}_{\mu\nu}(\mathbf{r}, \mathbf{r}'; \tau)$ of the coupled system (molecule and radiation), and calculate its new value by second-order perturbation theory (the first-order change vanishes).

We obtain the stationary states of the coupled system by the adiabatic theorem (Goldstone 1957; Gell-Mann & Low 1951; Bohm 1951). This states that if a perturbation V is switched on infinitely slowly between the times $t = -\infty$ and $t = 0$ each state of the unperturbed system goes smoothly into a corresponding stationary state of the perturbed system.

Let us consider how the density operator ρ' in the interaction representation changes from some initial value ρ_0 at $t = -\infty$ as we switch on V with a time factor $\mathrm{e}^{\alpha t}$. Eventually we go to the limit $\alpha = 0$, and $\rho'(0)$ tends to a certain limiting density $\hat{\rho}$ of the corresponding perturbed stationary states. The equation of motion

$$\mathrm{i}\hbar\,\mathrm{d}\rho'/\mathrm{d}t = -[\rho', V(t)]\,\mathrm{e}^{\alpha t};\ \rho'(t) = \mathrm{e}^{\mathrm{i}H_0 t/\hbar}\rho(t)\,\mathrm{e}^{-\mathrm{i}H_0 t/\hbar} \tag{5·1}$$

can be solved iteratively to give a series for $(\hat{\rho} - \rho_0)$ (Kubo & Tomita 1954) of which the first two terms are

$$\left.\begin{aligned} \rho^{(1)} &= \frac{\mathrm{i}}{\hbar}\int_{-\infty}^{0} [\rho_0, V(t_1)]\,\mathrm{e}^{\alpha t_1}\,\mathrm{d}t_1, \\ \rho^{(2)} &= \left(\frac{\mathrm{i}}{\hbar}\right)^2 \int_{-\infty}^{0}\int_{-\infty}^{t_1} [[\rho_0, V(t_2)], V(t_1)]\,\mathrm{e}^{\alpha(t_1+t_2)}\,\mathrm{d}t_1\,\mathrm{d}t_2. \end{aligned}\right\} \tag{5·2}$$

The value of the operator A at $t = 0$ is given by $\mathrm{Tr}\{\hat{\rho}A\}$, but it can also be written as the average over the unperturbed density ρ_0, of a new operator $\hat{A}$, such that $\mathrm{Tr}\{\hat{\rho}A\} = \mathrm{Tr}\{\rho_0\hat{A}\}$. Since this is to hold for arbitrary ρ_0 it is really an operator relation between the matrix elements of A in the old and new stationary states, and the relations $\mathrm{Tr}\{\rho^{(n)}A\} = \mathrm{Tr}\{\rho_0 A^{(n)}\}$ define a series expansion of $(\hat{A} - A)$ which we obtain from (5·2):

$$A^{(n)} = \left(\frac{-\mathrm{i}}{\hbar}\right)^n \int \mathrm{d}t_1 \int_{0>t_1>t_2\ldots>-\infty} \mathrm{d}t_2 \ldots \int \mathrm{d}t_n [[[A(0), V(t_1)], V(t_2)], \ldots V(t_n)]\,\mathrm{e}^{\alpha(t_1+t_2+\ldots t_n)}. \tag{5·3}$$

Notice that the order of operators $V(t_1)$, $V(t_2)$, $V(t_3)$ is reversed in going from (5·2) to (5·3). To illustrate this result, we put $V = -\int J_\nu(\mathbf{r}')\,A_\nu(\mathbf{r}')\,\mathrm{d}\mathbf{r}'$ and use (5·3) to calculate the first-order correction to the vector potential operator in the presence of a molecule

$$A_\mu^{(1)}(\mathbf{r}, 0) = \int_{-\infty}^{0} \mathrm{d}t' \int \mathrm{d}\mathbf{r}' \left\{\frac{\mathrm{i}}{\hbar}[A_\mu(\mathbf{r}, 0), A_\nu(\mathbf{r}', t')]\right\} J_\nu(\mathbf{r}', t')\,\mathrm{e}^{\alpha t'}, \tag{5·4}$$

which is just the retarded potential produced by the unperturbed currents (Schweber 1961).

At any finite time $t = -\tau$ we may regard the interaction $V\,\mathrm{e}^{-\alpha\tau}$ as being practically completely switched on, so that the behaviour of the partially coupled system between $-\tau$ and 0 will, as $\alpha \to 0$, become identical with that of the real system. If therefore a perturbation $F(t) = -A_\nu(\mathbf{r}', t)\,\delta(t+\tau)$ is superimposed on $V\,\mathrm{e}^{\alpha t}$ the change in $A_\mu(\mathbf{r}, 0)$ will measure the response function $\hat{K}$ of the coupled system.

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To calculate $\hat{K}$ we substitute $(V+F)$ for V in the expansion (5·3), keep terms linear in F, and take the average value over the unperturbed ground state. The replacement of V by F in $A^{(1)}$ gives the free field susceptibility K (2·18). The second-order correction $K^{(2)}$ comes from $A^{(3)}$ in (5·3) and is made up of three integrals derived from the triple commutators $[[[A,V],V],F]$, $[[[A,V],F],V]$ and $[[[A,F],V],V]$. Now A commutes with J, and the commutator (2·18) of the potentials is a pure number. These properties make all but the first integral vanish, and its value is

$$\left(\frac{\mathrm{i}}{\hbar}\right)^3 \int_{-\tau}^{0} \mathrm{d}t_1 \int_{-\tau}^{t_1} \mathrm{d}t_2 [[[A_\mu(\mathbf{r},0), V(t_1)], V(t_2)], A_\nu(\mathbf{r}',-\tau)]\, \mathrm{e}^{\alpha(t_1+t_2)} \tag{5·5}$$

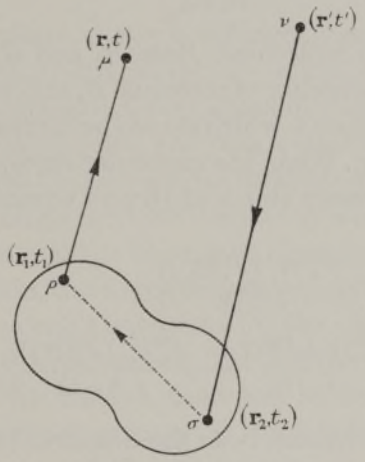


FIGURE 3

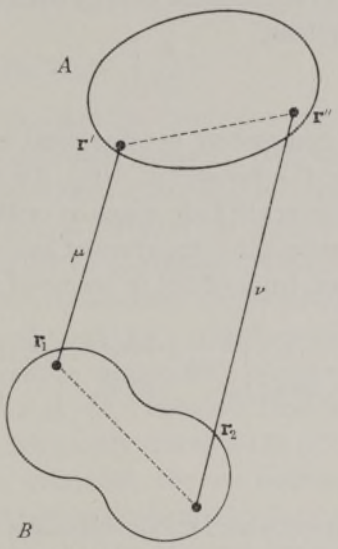


FIGURE 4

which, upon simplification yields the important equation

$$K^{(2)}_{\mu\nu}(\mathbf{r},\mathbf{r}';t-t') = \int_{t'}^{t} \mathrm{d}t_1 \int_{t'}^{t_1} \mathrm{d}t_2\, K_{\mu\rho}(\mathbf{r},\mathbf{r}_1;t-t_1) \left\langle \frac{\mathrm{i}}{\hbar}[J_\rho(\mathbf{r}_1,t_1), J_\sigma(\mathbf{r}_2,t_2)] \right\rangle K_{\sigma\nu}(\mathbf{r}_2,\mathbf{r}';t_2-t'). \tag{5·6}$$

Here we have relabelled the times and must finally integrate over $\mathbf{r}_1$ and $\mathbf{r}_2$. The commutator $[J,J]$ in (5·6) just gives the response function of the molecule, so that $K^{(2)}$ has the following significance (figure 3). A perturbation at $(\mathbf{r}',t')$ polarizes the radiation field, and produces a retarded potential at $(\mathbf{r}_2,t_2)$ which perturbs the molecule. The current thus set up at $(\mathbf{r}_1,t_1)$ produces in turn a retarded potential at $(\mathbf{r},\mathrm{t})$. The change of susceptibility is specially simple, as the Fourier transform of (5·6) gives the product

$$A^{(2)}_{\mu\nu}(\mathbf{r},\mathbf{r}';\omega) = \iint A_{\mu\rho}(\mathbf{r},\mathbf{r}_1;\omega)\, \alpha_{\rho\sigma}(\mathbf{r}_1,\mathbf{r}_2;\omega)\, A_{\sigma\nu}(\mathbf{r}_2,\mathbf{r}';\omega)\, \mathrm{d}\mathbf{r}_1\, \mathrm{d}\mathbf{r}_2 \tag{5·7}$$

at all frequencies, real or imaginary. In the dipole approximation with the reduced Hamiltonian (4·5) a precisely similar proof leads to

$$E^{(2)}_{ik}(\mathbf{r},\mathbf{r}';\omega) = E_{ij}(\mathbf{r},\mathbf{x};\omega)\, \alpha_{jl}(\omega)\, E_{lk}(\mathbf{x},\mathbf{r}';\omega) \tag{5·8}$$

for the electric field susceptibility.

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6. The force between two molecules

When one molecule (B) is in empty space it alters the susceptibility of the radiation field around it. If another molecule (A) is near, the self energy of A in the field changes, and causes a force between the molecules. Let $A^B_{\mu\nu}(\mathbf{r}', \mathbf{r}''; \omega)$ be the change of the field susceptibility in the presence of molecule B. Then the self-energy of A changes by

$$W = -\frac{\hbar}{2\pi}\int_0^\infty \mathrm{d}\xi \iint \alpha_{\mu\nu}(\mathbf{r}', \mathbf{r}''; \mathrm{i}\xi)\, A^B_{\nu\mu}(\mathbf{r}'', \mathbf{r}'; \mathrm{i}\xi)\, \mathrm{d}\mathbf{r}'\, \mathrm{d}\mathbf{r}''. \tag{6·1}$$

From (5·7) the mutual energy shift (up to the fourth order of perturbation theory) can be written

$$-\frac{\hbar}{2\pi}\int_0^\infty \alpha_{\mu\nu}(\mathbf{r}', \mathbf{r}''; \mathrm{i}\xi)\, \beta_{\nu\mu}(\mathbf{r}_2, \mathbf{r}_1; \mathrm{i}\xi)\, \frac{\mathrm{e}^{-k(R_1+R_2)}}{R_1 R_2}\, \mathrm{d}\xi \tag{6·2}$$

which is completely symmetrical in the two molecules. Here R_1 and R_2 are the distances $|\mathbf{r}' - \mathbf{r}_1|$ and $|\mathbf{r}'' - \mathbf{r}_2|$, β is the susceptibility of molecule B, and we should integrate over the four space co-ordinates. Figure 4, which should be compared with figure 3, shows the structure of this expression. When the molecular susceptibilities are written from (2·28) in terms of the stationary states of Dirac's equation, (6·2) becomes

$$-\frac{1}{2\pi\hbar c}\int_0^\infty \mathrm{d}k\, \frac{\mathrm{e}^{-k(R_1+R_2)}}{R_1 R_2}\left\{\frac{J^{0a}_\mu(\mathbf{r}')\, J^{a0}_\nu(\mathbf{r}'')}{a - \mathrm{i}k} + \frac{J^{0a}_\nu(\mathbf{r}'')\, J^{a0}_\mu(\mathbf{r}')}{a + \mathrm{i}k}\right\} \times \left\{\frac{J^{0b}_\nu(\mathbf{r}_1)\, J^{b0}_\mu(\mathbf{r}_2)}{b - \mathrm{i}k} + \frac{J^{0b}_\mu(\mathbf{r}_2)\, J^{b0}_\nu(\mathbf{r}_1)}{b + \mathrm{i}k}\right\}, \tag{6·3}$$

which agrees with Dzyaloshinskii's (1957) derivation from the S-matrix ($\hbar ca$ is the excitation energy of A). It is not convenient to pursue this formidable expression. Instead we shall look at the long and short-range limits of the energy in the dipole approximation with an effective Hamiltonian. The counterpart of (6·2) is

$$W = -\frac{\hbar}{2\pi}\int_0^\infty \mathrm{d}\xi \alpha_{ij}(\mathrm{i}\xi)\, E_{jk}(\mathbf{a}, \mathbf{b}; \mathrm{i}\xi)\, \beta_{kl}(\mathrm{i}\xi)\, E_{li}(\mathbf{b}, \mathbf{a}; \mathrm{i}\xi) \tag{6·4}$$

or

$$-\frac{\hbar}{2\pi}\int_0^\infty \mathrm{d}\xi \alpha_{ij}(\mathrm{i}\xi)\, \beta_{kl}(\mathrm{i}\xi)\left\{[\nabla_j \nabla_k - \delta_{jk}\nabla^2]\frac{\mathrm{e}^{-kR}}{R}\right\}\left\{[\nabla_l \nabla_i - \delta_{li}\nabla^2]\frac{\mathrm{e}^{-kR}}{R}\right\}. \tag{6·5}$$

The exponential ensures a simple limit at short distances. We replace e^{-kR} by 1 and use the standard formula for the electric polarizability α to obtain

$$W = -\sum_{a,b}\frac{1}{\hbar c(a+b)}\left|\frac{\mathbf{p}^{0a}.\mathbf{p}^{0b}}{R^3} - \frac{3(\mathbf{p}^{0a}.\mathbf{R})\,(\mathbf{p}^{0b}.\mathbf{R})}{R^5}\right|^2 \tag{6·6}$$

which is London's result. The long-range limit again involves the static polarizabilities $\alpha(0)$ and $\beta(0)$, since (6·5) reduces to

$$-\frac{\hbar c}{2\pi}\alpha_{ij}(0)\, \beta_{kl}(0)\int_0^\infty \mathrm{d}k \left\{[\nabla_j \nabla_k - \delta_{jk}\nabla^2]\frac{\mathrm{e}^{-kR}}{R}\right\}\left\{[\nabla_l \nabla_i - \delta_{li}\nabla^2]\frac{\mathrm{e}^{-kR}}{R}\right\}. \tag{6·7}$$

The physical interpretation of (6·7) is obvious if we go back to (6·1). The coefficient of $\beta_{kl}(0)$,

$$-\frac{\hbar}{2\pi}\int_0^\infty E^A_{lk}(\mathbf{b}, \mathbf{b}; \mathrm{i}\xi)\, \mathrm{d}\xi = -\tfrac{1}{2}\langle E_l(\mathbf{b})\, E_k(\mathbf{b})\rangle^A, \tag{6·8}$$

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is the mean square electric field produced at B by molecule A, and W is the polarization energy of B in this field,

$$W = -\tfrac{1}{2}\beta_{kl}\langle E_l E_k\rangle^A. \tag{6·9}$$

To evaluate the integral in (6·7) we will take the line joining the molecules to be the x axis, use the expressions (4·15) for the components of E_{ik}, and assume that the x, y, z axes are principal axes of the polarizability tensors. The interaction energy is

$$-\frac{\hbar c}{8\pi R^7}\{20\alpha_{xx}(0)\beta_{xx}(0) + 13[\alpha_{yy}(0)\beta_{yy}(0) + \alpha_{zz}(0)\beta_{zz}(0)]\} \tag{6·10}$$

for molecules, or $-[23\hbar c/4\pi R^7]\alpha(0)\beta(0)$ for atoms.

7. Discussion

The present theory is sufficiently general to calculate the dispersion energy of two molecules of any shape by fourth order perturbation theory. One may easily include magnetic dipole, spin, electric quadrupole, and other higher terms if they are ever important. The very weak fluctuation forces between two atoms which are coupled to the zero-point vibrations of a crystal lattice (Grimley 1962) can be dealt with similarly, and it is easy to generalize the theory to finite temperatures.

We derived the important equations (2·19) and (2·26) by a quantum-mechanical calculation, but a simple classical argument gives the same result. Suppose that we solve Maxwell's equations for the electric field $E_i(\mathbf{r})$ of a classical Hertzian dipole $P_k e^{\xi t}$ at $\mathbf{r}'$ in free space. The field turns out to be exactly the same as in (4·10) and (4·15). This follows from the correspondence principle, for we notice that the response function involves the commutator $(\mathrm{i}/\hbar)\,[E_i(\mathbf{r}, t), E_k(\mathbf{r}', t')]$. Now according to Dirac (1958) the commutator is proportional to the Poisson bracket of the classical quantities. As the Poisson brackets determine the response of the classical system, we see that the susceptibility of the electromagnetic field must come out exactly the same in quantum or classical mechanics. If one also deduces equation (5·8) from a classical argument, then the only quantum-mechanical proof which is absolutely necessary to establish Casimir & Polder's result is equation (3·2), which is merely a transformation of the standard formula of second-order perturbation theory.

When there are only a few molecules the perturbation series converges rapidly and the dispersion forces are nearly additive. On the other hand an electromagnetic wave can be reflected many times as it travels between the surfaces of two macroscopic bodies, and a perturbation calculation must be carried to infinite order. Hence the non-additivity and the necessity for Lifschitz's (1956) general theory.

I would like to thank Professor H. C. Longuet-Higgins, F.R.S., for suggesting this problem and for much valuable advice, and Dr M. J. Stephen for a very helpful discussion.

Appendix. The reduced Hamiltonian

In a neutral molecule the charge ρ can be replaced by a certain density of electric dipoles $\mathbf{P}$, the 'electric displacement', such that $\rho = -\operatorname{div}\mathbf{P}$; and the current, $\mathbf{J} = \operatorname{curl}\mathbf{M} + (1/c)\dot{\mathbf{P}}$ consists of the displacement current and a current $\operatorname{curl}\mathbf{M}$ which

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produces no movement of charge, and may, for example, arise from electron spin. The interaction term in the Lagrangian of an electron and radiation now becomes

$$\mathscr{L}_I = -\int (\rho\phi - \mathbf{J}.\mathbf{A})\,\mathrm{d}v = -\int \left\{-\phi\,\mathrm{div}\,\mathbf{P} - \frac{1}{c}\dot{\mathbf{P}}.\mathbf{A} - \mathbf{A}.\mathrm{curl}\,\mathbf{M}\right\}\mathrm{d}v$$

$$= \int \{\mathbf{P}.\mathbf{E} + \mathbf{M}.\mathbf{H}\}\,\mathrm{d}v + \frac{1}{c}\frac{\mathrm{d}}{\mathrm{d}t}\int \mathbf{P}.\mathbf{A}\,\mathrm{d}v \qquad (\mathrm{A}\,1)$$

after the integrals have been transformed by Green's theorem. (We assume that $\mathbf{P}, \mathbf{M}, \mathbf{A}$ and ϕ each vanish at infinity, so that all surface integrals in the transformation vanish.) The last term of (A 1) is the time derivative of a quantity $\mathbf{P}.\mathbf{A}$ which does not depend explicitly on the time, and it produces no physical effects. To see this, consider first a classical system, and suppose that the Lagrangian $\mathscr{L}$ is changed to

$$\mathscr{L}' = \mathscr{L}(q, \dot{q}, t) + \frac{\mathrm{d}F(q)}{\mathrm{d}t} = \mathscr{L}(q, \dot{q}, t) + \frac{\partial F}{\partial q}\dot{q}. \qquad (\mathrm{A}\,2)$$

Then the change in the action $S = \int \mathscr{L}\,\mathrm{d}t$ for any path from a to b is $F(b) - F(a)$, which does not depend on the path. The equations of motion are therefore unchanged. In quantum mechanics the change from $\mathscr{L}$ to $\mathscr{L}'$ also leaves the operator equations of motion unchanged. But since the momentum changes to

$$p' = (p + \partial F/\partial q),$$

the Hamiltonian $\mathscr{H}' = (p'\dot{q} - \mathscr{L}')$ takes a new form. However, each stationary state ψ' of $\mathscr{H}'$ is related to the corresponding state ψ of $\mathscr{H}$ by the unitary transformation $\psi' = \mathrm{e}^{-\mathrm{i}F/\hbar}\psi$ and has the same energy.

To prove that the interaction Hamiltonian $\mathscr{H}_I = -\mathscr{L}_I$ changes to

$$\mathscr{H}'_I = -\int \{\mathbf{P}.\mathbf{E} + \mathbf{M}.\mathbf{H}\}\,\mathrm{d}v \qquad (\mathrm{A}\,3)$$

when the $\mathbf{P}.\mathbf{A}$ term is dropped from (A 1) we adapt Power & Zienau's argument (1957).

For simplicity we consider a one-dimensional neutral atom (one electron) interacting with radiation, since this problem illustrates the essential steps of the argument. The original Lagrangian $\mathscr{L}$ is the sum of

$$\mathscr{L}_A = \tfrac{1}{2}m\dot{q}^2 - V(q), \quad \mathscr{L}_R = \frac{1}{2}\left\{\left(\frac{\partial A}{\partial t}\right)^2 - \left(\frac{\partial A}{\partial x}\right)^2\right\},$$

$$\mathscr{L}_I = e\{\phi(q) - \phi(0) - A\dot{q}\} = e\left\{q\frac{\partial \phi}{\partial x} - A\dot{q}\right\};$$

for the atom, the radiation, and the interaction. $\mathscr{L}_R$ should also contain terms in ϕ, but these play no essential part in the present argument. We have also changed temporarily to a set of units in which $\hbar = c = 1$. The momenta conjugate to q and A are $p = (m\dot{q} - eA)$, $B = \dot{A}$, and we immediately obtain the standard Hamiltonian

$$\mathscr{H} = [(p + eA)^2/2m + V] - eq\frac{\partial \phi}{\partial x} + \frac{1}{2}\left[B^2 + \left(\frac{\partial A}{\partial x}\right)^2\right], \qquad (\mathrm{A}\,4)$$

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where the interaction term is $\mathscr{H}_I = [-JA - eq\nabla\phi]$. To find the reduced Hamiltonian we add $e(A\dot{q}+q\dot{A})$ to $\mathscr{L}_I$, which changes into $\mathscr{L}'_I = -eq(-\nabla\phi - \dot{A}) = -eqE$. The new momenta are $p' = m\dot{q}$, $B' = (\dot{A}+eq)$, and give the new Hamiltonian function

$$\mathscr{H}' = \left[\frac{1}{2m}p'^2 + V(q)\right] - eq\frac{\partial\phi}{\partial x} + \frac{1}{2}\left[(B'-eq)^2 + \left(\frac{\partial A}{\partial x}\right)^2\right] \tag{A 5}$$

in which the interaction part $\mathscr{H}'_I = -eq[\nabla\phi + (B'-eq)]$ is indeed the same as the reduced form $\mathscr{H}'_I = eqE$.

The derivation of (A 3) does not assume that the molecule is small compared with the wavelength of light. If it is, then $\mathbf{P}(\mathbf{r})$ and $\mathbf{M}(\mathbf{r})$ reduce to point dipoles and the effective Hamiltonian for a molecule at $\mathbf{r}$ becomes

$$\mathscr{H}'_I = -\mathbf{P}.\mathbf{E}(\mathbf{r}) - \mathbf{M}.\mathbf{H}(\mathbf{r}). \tag{A 6}$$

When the dipole moment of a transition vanishes it may be necessary to consider the electric quadrupole tensor $\mathbf{Q}$. In this case the electric polarization in (A 6) is replaced by $\mathbf{P}(\mathbf{r}) = \operatorname{div}\mathbf{Q}(\mathbf{r})$, in which $\mathbf{Q}(\mathbf{r})$ is the density of quadrupole moment. The resulting space integral is transformed by Green's theorem, and gives the effective Hamiltonian $\mathscr{H}'_I = \mathbf{Q}.\nabla\mathbf{E}(\mathbf{r})$ for a point quadrupole.

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